
Course Syllabus

Real-Time Operating Systems

(Rabb patibatkan baeb Real-time)

Course Information

Course Code	TBD
Course Title (EN)	Real-Time Operating Systems
Course Title (TH)	Rabb Patibatkan Baeb Real-time (see Thai version)
Credits	3 Credits (3–0–6)
Level	Master of Engineering (M.Eng.)
Program	Electrical and Computer Engineering
Faculty	Faculty of Engineering, KMUTNB
Semester	<i>[Semester / Academic Year]</i>
Instructor	<i>[Instructor Name]</i> , Ph.D.
Contact	Department of ECE, KMUTNB
Office Hours	By appointment

1. Course Description

This course provides a rigorous graduate-level treatment of real-time operating systems (RTOS), covering both theoretical foundations and practical implementation on modern embedded hardware. Topics include real-time scheduling theory, task synchronization, inter-process communication, memory management, timing analysis, worst-case execution time (WCET) estimation, and advanced architectural features including TrustZone-M security partitioning and multi-core asymmetric multiprocessing (AMP).

Laboratory sessions are conducted on the **NXP FRDM-MCXXN236** development board (dual Cortex-M33 at 150 MHz, 512 KB SRAM) using **FreeRTOS** as the primary RTOS, with a comparative module on **Zephyr RTOS**. Students gain hands-on experience with professional-grade tools including SEGGER SystemView for real-time task tracing and the NXP MCUXpresso SDK.

2. Prerequisites

- Embedded systems or microcontroller programming (undergraduate level)
- C programming proficiency
- Basic operating systems concepts
- Familiarity with ARM Cortex-M architecture is beneficial but not required

3. Course Objectives (CLOs)

Upon successful completion of this course, students will be able to:

- CLO1. Explain** the theoretical foundations of real-time scheduling algorithms including Rate Monotonic Scheduling (RMS) and Earliest Deadline First (EDF), and perform schedulability analysis.
- CLO2. Analyze** priority inversion problems and apply priority inheritance and priority ceiling protocols to resolve resource contention.
- CLO3. Implement** multi-task RTOS applications using FreeRTOS on ARM Cortex-M33, including task management, queues, semaphores, mutexes, and event groups.
- CLO4. Measure** worst-case execution time (WCET) and interrupt latency using hardware cycle counters (DWT) and SEGGER SystemView tracing.
- CLO5. Design** secure embedded systems leveraging TrustZone-M for secure/non-secure world partitioning.
- CLO6. Develop** multi-core AMP applications utilizing dual Cortex-M33 cores with inter-core communication via shared memory and mailbox.
- CLO7. Compare** RTOS design philosophies by contrasting FreeRTOS and Zephyr RTOS in terms of architecture, API, and toolchain.
- CLO8. Evaluate** current research literature in real-time systems and apply findings to practical embedded system design.

4. Hardware and Software Tools

4.1. Hardware Platform

Development Board	NXP FRDM-MCXN236
MCU	NXP MCX N236 (dual Cortex-M33 @ 150 MHz)
Memory	1 MB Flash, 512 KB SRAM
Security	TrustZone-M, EdgeLock Secure Subsystem
NPU	eIQ Neutron Neural Processing Unit
Debugger	Onboard MCU-Link (re-flashable to J-Link firmware)

4.2. Software Tools

IDE	NXP MCUXpresso IDE / VS Code + MCUXpresso Extension
SDK	NXP MCUXpresso SDK (FreeRTOS included)
Debugger FW	MCU-Link re-flashed to J-Link (SEGGER, free)
Tracer	SEGGER SystemView (free educational license)
Debugger IDE	SEGGER Ozone (free with J-Link)
Secondary RTOS	Zephyr RTOS (west build system)

5. Required Texts and References

5.1. Required

1. **Laplace, P.A.** *Real-Time Systems Design and Analysis: Tools for the Practitioner*, 4th ed., IEEE Press/Wiley. [Main theory text]
2. **Barry, R.** *Mastering the FreeRTOS Real Time Kernel*. FreeRTOS.org. [Free PDF at freertos.org]
3. **Yiu, J.** *The Definitive Guide to Arm Cortex-M23 and Cortex-M33 Processors*. Newnes/Elsevier, 2021.

5.2. Seminal Papers (via KMUTNB IEEE Xplore Access)

1. **Liu & Layland (1973)** – “Scheduling Algorithms for Multiprogramming in a Hard Real-Time Environment,” *J. ACM*, 20(1):46–61.
2. **Sha, Rajkumar & Lehoczky (1990)** – “Priority Inheritance Protocols: An Approach to Real-Time Synchronization,” *IEEE Trans. Computers*, 39(9).
3. **Wilhelm et al. (2008)** – “The Worst-Case Execution-Time Problem: Overview of Methods and Survey of Tools,” *ACM TECS*, 7(3).

5.3. Supplementary

1. **Valvano, J.W.** *Embedded Systems: RTOS for ARM Cortex-M*. [Free at users.ece.utexas.edu/~valvano]
2. NXP AN13958 – Getting Started with FreeRTOS on MCX N Series.
3. NXP AN13813 – TrustZone-M on MCX N Series.
4. MCX N236 Reference Manual, NXP Semiconductors.
5. SEGGER SystemView User Guide, SEGGER Microcontroller GmbH.

6. Course Schedule

Week	Lecture Topics	Lab / Activity	Reading
Module 1: Foundations of Real-Time Systems			
1	RT system taxonomy; hard/soft/firm dead-lines; RTOS vs. GPOS characteristics	Toolchain setup: MCUXpresso IDE, MCU-Link to J-Link re-flash, SystemView install	Laplace Ch. 1–2 Barry Ch. 1
2	Task model; periodic, aperiodic, sporadic; timing constraints; utilization bound	Lab 1: Hello FreeRTOS — task creation, priorities, <code>vTaskDelay</code> , first SystemView trace	Laplace Ch. 3 Liu & Layland (1973)
Module 2: Real-Time Scheduling Theory			

Week	Lecture Topics	Lab / Activity	Reading
3	Rate Monotonic Scheduling (RMS); schedulability analysis; Liu & Layland theorem	Lab 2: Fixed-priority preemptive scheduling; observe context switches in SystemView	Laplane Ch. 5 Liu & Layland (1973)
4	Earliest Deadline First (EDF); dynamic priority; RMS vs. EDF comparison; hyperperiod	Paper discussion: Liu & Layland (1973) — schedulability proofs	Laplane Ch. 6 Barry Ch. 3
5	Aperiodic and sporadic task handling; polling server; deferrable server; sporadic server	Lab 3: Aperiodic handling with FreeRTOS software timers and deferred ISR pattern	Laplane Ch. 6 Barry Ch. 7
Module 3: Synchronization and Inter-Task Communication			
6	Queues, semaphores, mutexes, event groups; FreeRTOS sync API; blocking behavior	Lab 4: Producer–consumer queue; binary semaphore ISR-to-task sync; event groups	Barry Ch. 6–8 Laplane Ch. 7
7	Priority inversion; Priority Inheritance Protocol (PIP); Priority Ceiling Protocol (PCP); Stack Resource Policy (SRP)	Lab 5: Priority inversion demo (3-task); resolve with FreeRTOS mutex; verify in SystemView	Sha et al. (1990) Laplane Ch. 7
8	Deadlock; livelock; starvation; watchdog timers; defensive RTOS design patterns	Paper discussion: Sha et al. (1990) — PIP/PCP; mid-term review	Laplane Ch. 8
Mid-Term Examination (after Week 8)			
Module 4: Timing Analysis and Memory Management			
9	WCET: definition, measurement, static analysis; DWT cycle counter on Cortex-M33; interrupt latency	Lab 6: WCET measurement using DWT; interrupt latency profiling; estimated vs. measured comparison	Wilhelm et al. (2008) Yiu Ch. 9, 14

Week	Lecture Topics	Lab / Activity	Reading
10	FreeRTOS memory management: heap_1–heap_5; stack sizing; overflow detection; fragmentation	Lab 7: Heap comparison (heap_4 vs. heap_5); <code>uxTaskGetStackHighWaterMark</code> ; overflow hook	Barry Ch. 2 Laplane Ch. 8
Module 5: Advanced Hardware Features (FRDM-MCXXN236)			
11	ARM TrustZone-M: secure/non-secure world; SAU configuration; secure gateway veneer; RTOS in TrustZone	Lab 8: TrustZone-M on MCXX N236 — FreeRTOS in non-secure world; secure peripheral access	Yiu Ch. 17–19 NXP AN13813
12	Multi-core AMP vs. SMP; inter-core communication; shared memory; MCXX N236 mailbox unit	Lab 9: Dual-core AMP — FreeRTOS on Core 0, bare-metal sensor on Core 1; mailbox data transfer	Yiu Ch. 20 MCXX N236 RM
Module 6: Power Management and Zephyr RTOS			
13	Low-power RTOS: tickless idle; <code>configUSE_TICKLESS_IDLE</code> ; Cortex-M33 sleep modes; power measurement	Lab 10: Tickless idle on FRDM-MCXXN236; measure active vs. idle current; scheduling optimization	Barry Ch. 10 Yiu Ch. 13
14	Zephyr RTOS: Kconfig, device tree, kernel services; comparison with FreeRTOS; west build	Lab 11: Port FreeRTOS application to Zephyr (<code>frdm_mcxn236</code> target); compare SystemView traces	Zephyr Docs Nordic Dev. Academy
Module 7: Research Topics and Final Project			
15	Research frontiers: mixed-criticality systems; AUTOSAR OS; RTOS+ML (eIQ NPU); safety standards IEC 61508, ISO 26262	Project milestone review; SystemView analysis consultation; thesis topic discussion	Recent IEEE RTSS / RTAS papers
16	Final Project Presentations — 15-minute live demonstration + Q&A per group		

7. Assessment and Grading

Assessment Component	Weight
Lab Reports (Labs 1–11, best 10 counted)	30%
Paper Discussions (Weeks 4 and 8)	10%
Mid-Term Examination	20%
Final Project (design + demo + report)	30%
Participation and in-class activities	10%
Total	100%

Grading Scale: A: $\geq 80\%$ B+: 75–79% B: 70–74% C+: 65–69% C: 60–64%
D/F: $< 60\%$

8. Final Project

Students (individually or in pairs) design and implement a complete real-time embedded application on the FRDM-MCXXN236 board. Suggested themes:

- **IoT sensor node** — multi-task FreeRTOS with UART/LoRa, tickless idle power optimization
- **Motor control RTOS** — hard real-time PWM task + supervisory tasks with schedulability report
- **Secure edge gateway** — TrustZone-M with secure key storage and non-secure FreeRTOS app
- **Dual-core AMP system** — sensor acquisition on Core 1, data processing + communication on Core 0
- **RTOS + ML inference** — FreeRTOS scheduler with eIQ NPU task; scheduling overhead analysis

Deliverables:

1. Final report including schedulability analysis, SystemView screenshots, and WCET measurements
2. Live hardware demonstration (Week 16)
3. Source code submission via university LMS

9. Laboratory Summary

Lab	Title	Key Skills Developed
1	Hello FreeRTOS	Task API, SystemView setup and configuration
2	Preemptive Scheduling	Priority assignment, context switch observation
3	Aperiodic Task Handling	Software timers, deferred interrupt service
4	IPC Mechanisms	Queues, semaphores, event groups
5	Priority Inversion	Mutex, PIP demonstration, SystemView analysis
6	WCET Measurement	DWT cycle counter, interrupt latency profiling
7	Memory Management	Heap scheme comparison, stack high-water mark
8	TrustZone-M Security	SAU config, secure/non-secure partitioning
9	Dual-Core AMP	Inter-core mailbox, multi-core RTOS design
10	Low-Power RTOS	Tickless idle, current measurement, optimization
11	Zephyr RTOS	West toolchain, device tree, API comparison

10. Course Policies

- **Lab attendance** is mandatory. Missed labs may be made up within one week with prior instructor notification.
- **Late submissions** are penalized 10% per calendar day.
- **Hardware responsibility:** Each student is responsible for the FRDM-MCXXN236 board assigned to them. Damage must be reported immediately.
- **Paper discussions** require a one-page critical summary submitted before class via LMS.
- All submitted work must represent the student's own effort in accordance with KMUTNB academic integrity regulations.

11. Reading Map by Course Module

Topic	Primary Source	Supplement
Scheduling theory (RMS, EDF)	Laplante Ch. 5–6	Liu & Layland (1973)
Priority inversion / PIP	Laplante Ch. 7	Sha et al. (1990)
WCET analysis	Wilhelm et al. (2008)	Yiu Ch. 14
FreeRTOS API	Barry Ch. 1–10	MCUXpresso SDK examples
Cortex-M33 internals	Yiu Ch. 1–16	MCX N236 Reference Manual
TrustZone-M	Yiu Ch. 17–19	NXP AN13813
Dual-core AMP	Yiu Ch. 20	MCX N236 RM (Mailbox)
Low-power design	Barry Ch. 10	Yiu Ch. 13
Zephyr RTOS	Zephyr Docs	Nordic Developer Academy
Research frontiers	IEEE RTSS/RTAS papers	Course handouts

This syllabus is subject to modification. Any changes will be announced via the course LMS and take effect no earlier than one week after notification.

Instructor Signature:

Instructor

Date